

Rapid Spanning Tree Protocol (RSTP):

Rapid Spanning Tree Protocol (RSTP), defined in IEEE 802.1w, is an evolution of the original Spanning Tree Protocol (STP) aimed at providing faster convergence times in network topology changes. RSTP significantly improves the speed at which a network can recover from failures, making it more suitable for modern, high-availability networks.

Key Features of RSTP

- 1. Faster Convergence:**
- 2. Port Roles:**
- 3. Port States:**
- 4. Edge Ports:**
- 5. Link Types:**

Port Roles in RSTP:

Root Port (RP): The port with the best path to the root bridge.

Designated Port (DP): The port designated to forward frames to/from a network segment.

Alternate Port: A backup port to the root port, providing an alternate path to the root bridge.

Backup Port: A backup port to the designated port, providing redundancy on a network segment.

Port States in RSTP:

Discarding: Combines the blocking and listening states of STP, where the port does not forward frames or learn MAC addresses.

Learning: The port learns MAC addresses but does not forward frames.

Forwarding: The port forwards frames and learns MAC addresses.

Port Roles in RSTP:

<https://www.youtube.com/@techwithravish>

BID: Switch 0
32768 +
0001.A123.B456
(Root Bridge)



DES



DES

ROOT



ROOT



BID: Switch 1
32768 +
0003.A123.B456



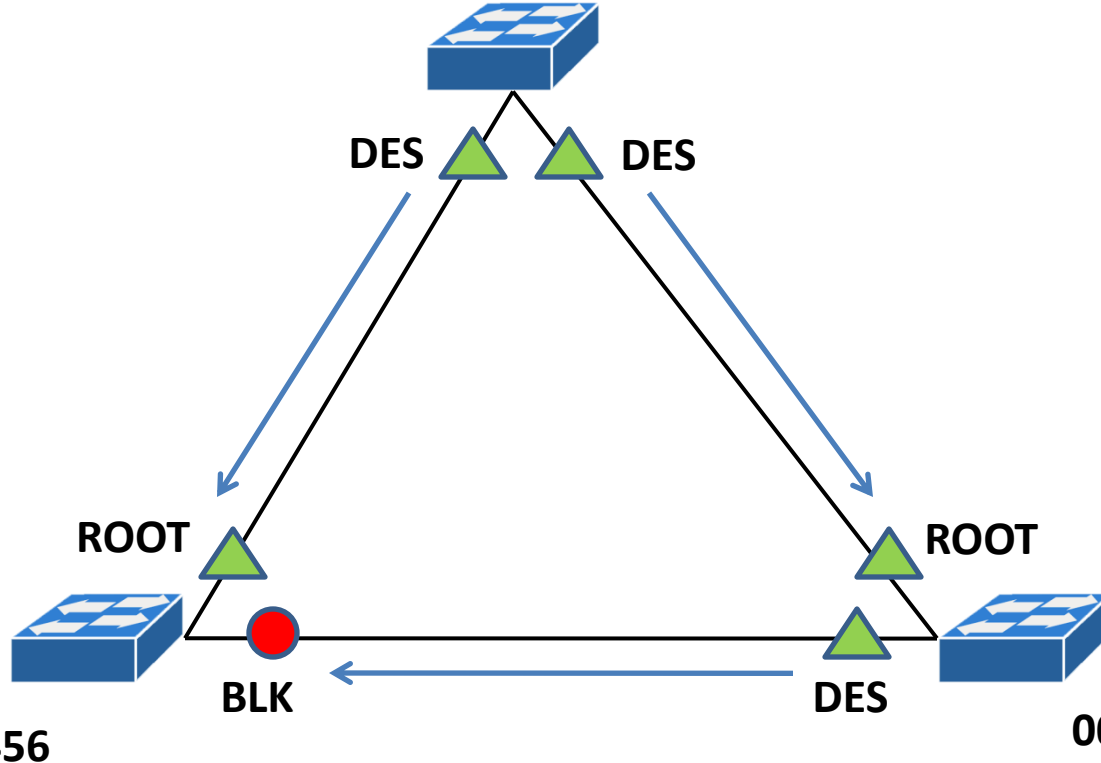
BLK



DES



BID: Switch 2
32768 +
0002.A123.B456

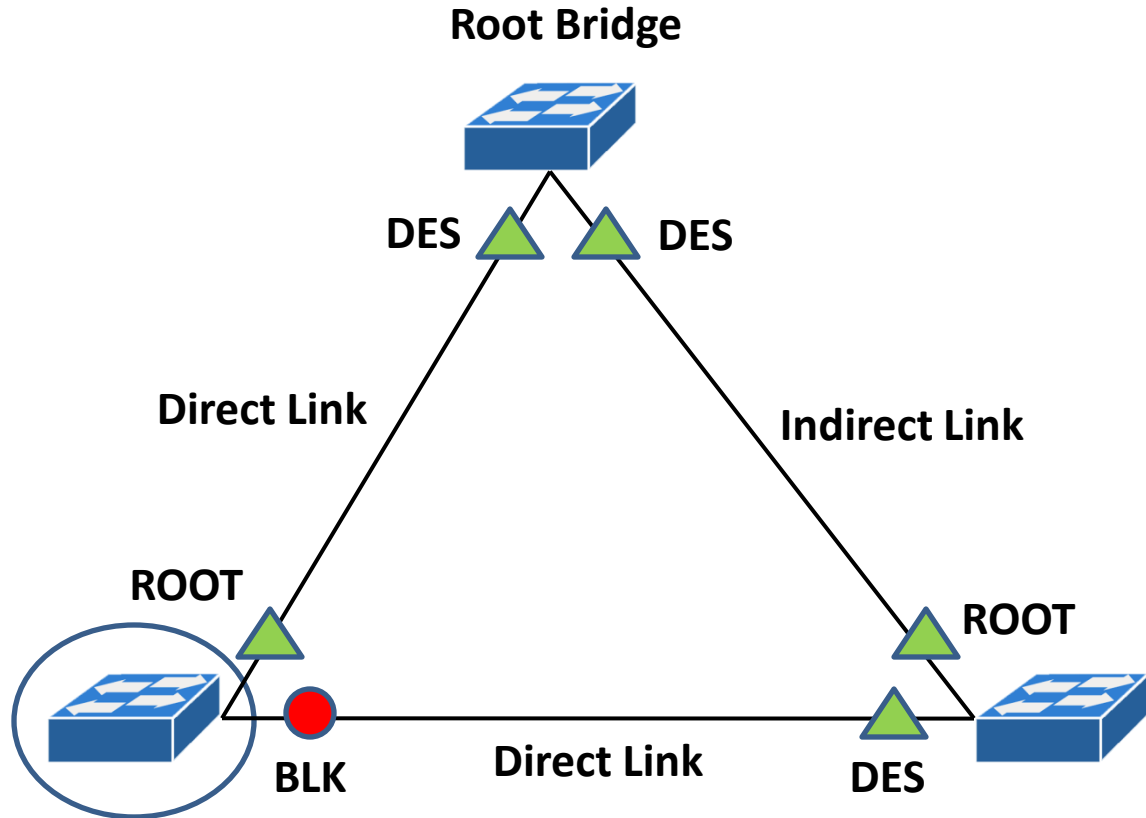


RSTP Links:

Direct Link: Link

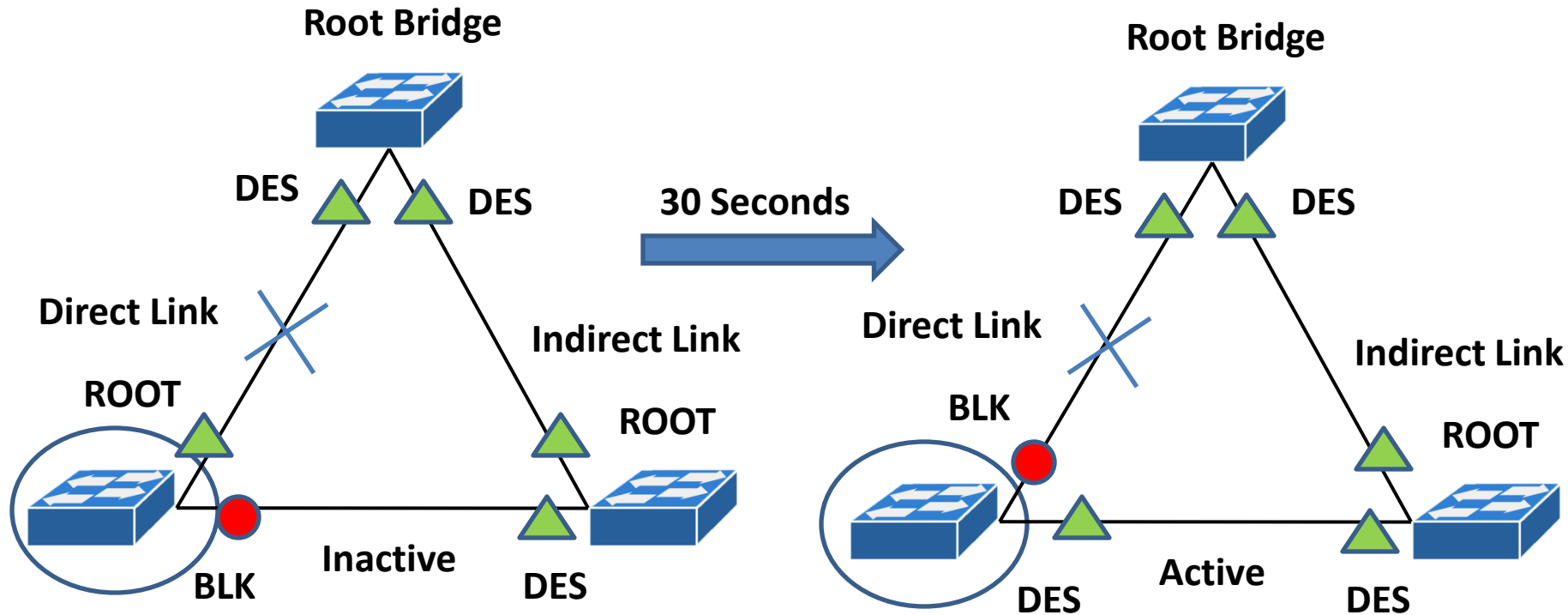
Adjacent or Directly Connected to Switch is called Direct Link.

Indirect Link: Link Not Adjacent or In Directly Connected to Switch is called Indirect Link.



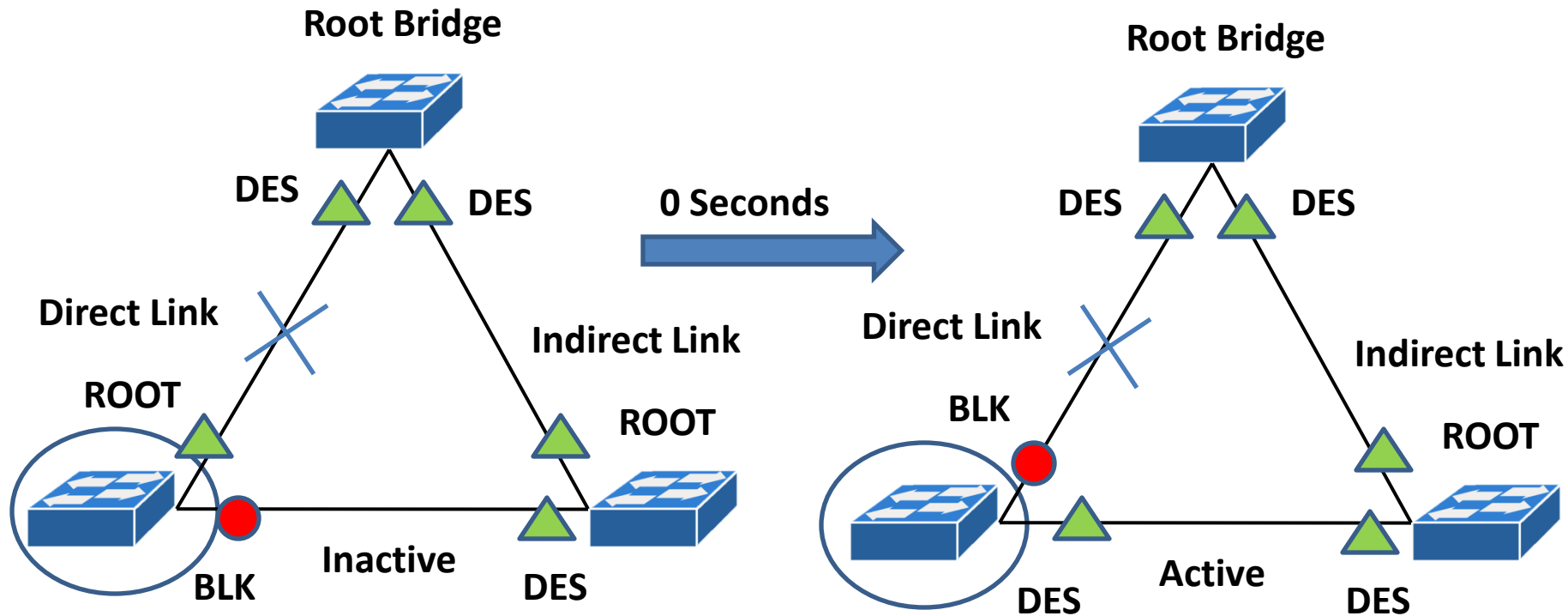
STP Convergence Time:

If Direct Link is Down: 15 Sec (LSN) + 15 Sec (LRN)= 30 Sec



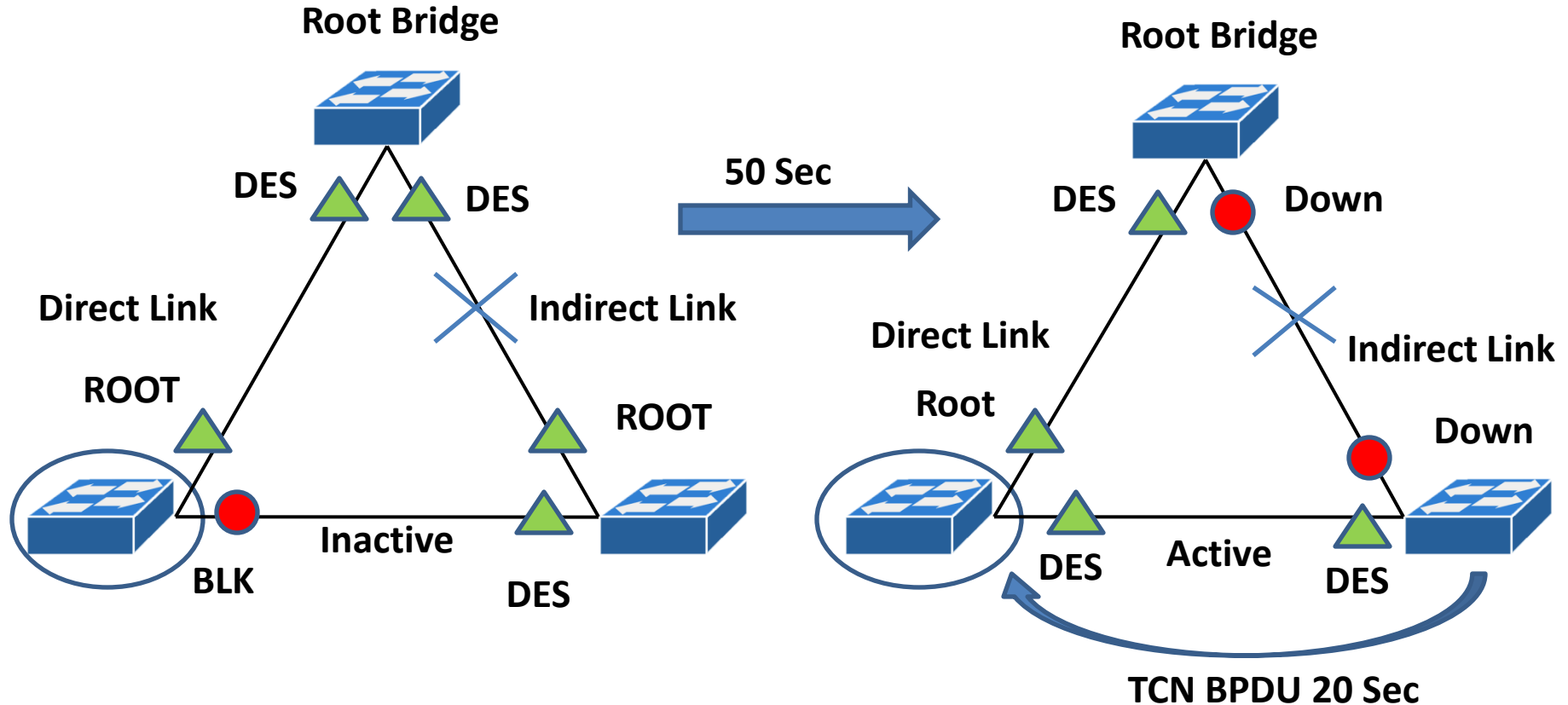
RSTP Convergence Time:

If Direct Link is Down: 0 Seconds



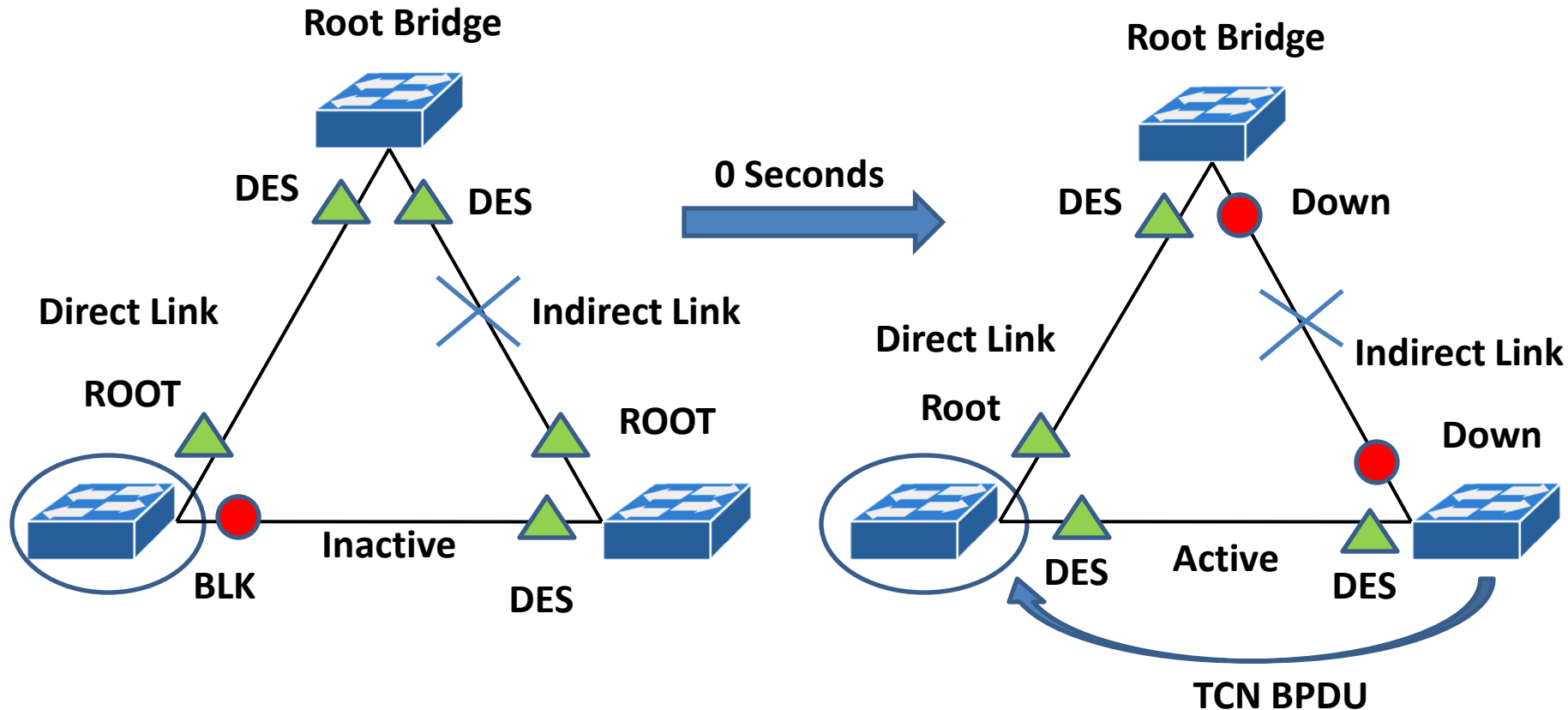
STP Convergence Time:

If Indirect Link Down: 15 Sec (LSN) + 15 Sec (LRN)+ Wait 20 Sec= 50 Sec



RSTP Convergence Time:

If Indirect Link Down: 0 Seconds



Configuration of RSTP:

RSTP Config:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# spanning-tree mode rapid-pvst
```

Portfast on Interfaces:

```
Switch(config)# interface fastethernet 0/1
```

```
Switch(config-if)# spanning-tree portfast
```

```
Switch(config-if)# end
```